Chest X-Ray Pneumonia Screening

MEDIMG FS26 — Decision Support in Medical Imaging

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Binary triage: **NORMAL vs PNEUMONIA** · Reproducible repo on GitHub

Motivation

- **Clinical need:** prioritize likely pneumonia cases in busy workflows
- **Our role:** decision **support** — not autonomous diagnosis
- **Success criteria:** ranking quality (AUC) + transparent error types (FP/FN) at a stated threshold

Dataset & Protocol

Item	Choice
Data	`hf-vision/chest-xray-pneumonia` (CC BY 4.0)
Train	~5.2k (HF train)
Validation	**10% stratified hold-out from train** (HF val ≈16 → too small)
Test	**624** HF test images — **never used for training**
Input	224×224 RGB, ImageNet normalization

Dataset (quick snapshot)

- **Task labels:** NORMAL vs PNEUMONIA
- **Test set label count:** 234 NORMAL · 390 PNEUMONIA
- **Why this dataset:** small, public, easy to reproduce end-to-end; good for studying baselines + ablations
- **Caveat:** single dataset → limited external validity (no hospital/site shift tested)

Evaluation

- **ROC-AUC** — threshold-free ranking
- **Accuracy & F1** — at fixed threshold **0.5** (not tuned for deployment)
- **Confusion matrix** — false positives on normals vs false negatives on pneumonia
- **Robustness:** seeds 0–2 for main models; ablations: aug, loss, 25% train data

Baselines (test set)

Method	AUC	Acc	F1
Random (train prevalence)	0.52	0.58	0.69
Majority class	0.50	0.63	0.77
Frozen ResNet18 + logistic regression	**0.95**	**0.84**	**0.89**

Takeaway: strong pretrained features already solve most of the task.

Model 1 — ResNet18 + cross-entropy

- ImageNet **ResNet18**, fully fine-tuned
- Loss: **cross-entropy**
- Main variant: **augmentation on** (flip, rotation, color jitter, **RandomErasing**)

Model 2 — ResNet18 + focal loss

- Same **ResNet18** backbone (fine-tuned)
- **Focal loss** ($\gamma = 2$) instead of CE — focus on hard examples
- Trained with augmentation on (same protocol as Model 1 aug)

Results — fine-tuned vs baseline (test AUC)

Model	AUC (seed 0)	AUC (seeds 0–2 mean)
Frozen ResNet18 + LR	0.949	—
CE, **no aug**	0.935	—
CE, **aug on**	0.960	**0.954**
Focal, **aug on**	0.957	**0.953**

Interpretation: gains over frozen LR are **small (~0.01–0.02 AUC)** — useful but not dramatic.

Results — threshold 0.5 matters (234 normals on test)

False positives = normal scans called pneumonia (cm_{01} / 234):

Run	FP count	Rate on normals
Frozen LR	94	40%
CE, no aug	108	46%
CE, aug (seed 0)	**140**	**60%**
CE, aug (seed 1)	72	31%
CE, aug (seed 2)	70	30%

Interpretation: higher AUC does **not** always mean fewer FPs at 0.5; aug changed **score calibration**, not a clear win on all metrics.

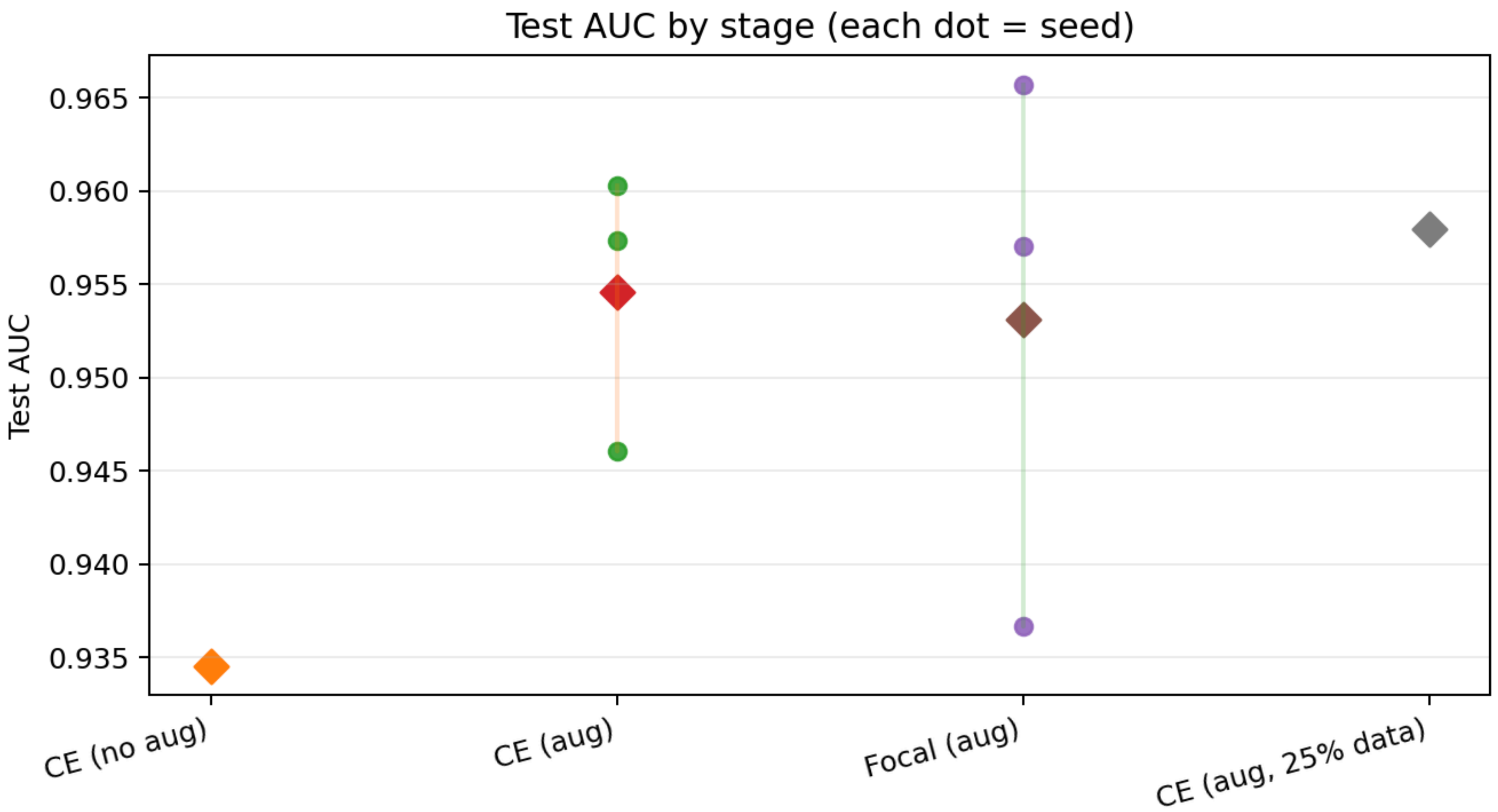
Ablations (seed 0, test)

Ablation	AUC	Acc	F1
Aug **off** (CE)	0.935	0.82	0.87
Aug **on** (CE)	0.960	0.78	0.85
Focal vs CE (aug on)	0.957 vs 0.960	~0.81 vs ~0.78	~0.87 vs ~0.85
25% train data (CE, aug)	0.958	0.85	0.89

25% run: **single seed** — illustrative only, not a strong claim.

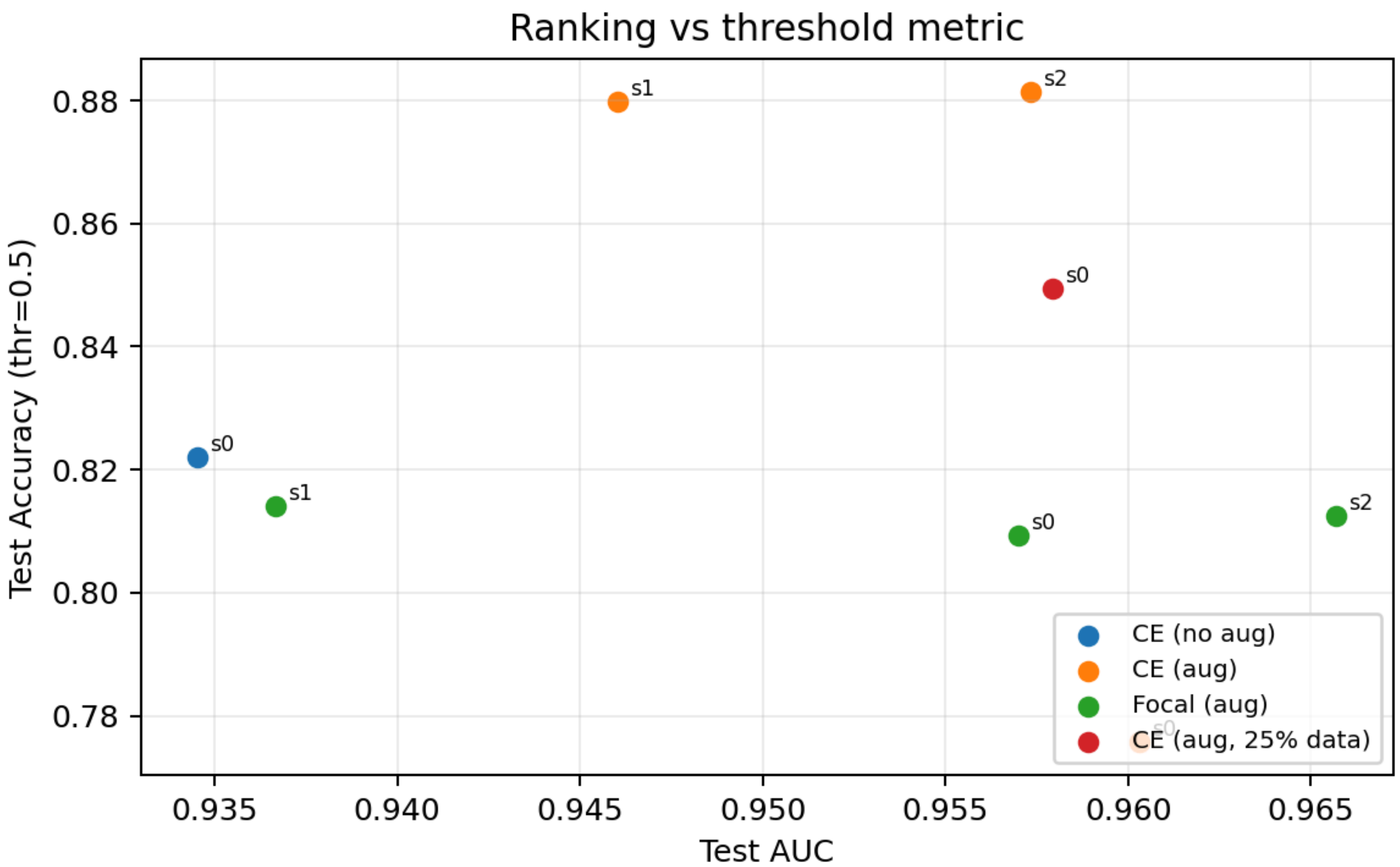
Summary plot — AUC across stages (seeds)

- AUC is **high** across runs; differences are **incremental**.
- CE-aug and focal-aug show **moderate** seed-to-seed variance.



Summary plot — AUC vs accuracy (thr=0.5)

- AUC measures **ranking**, accuracy depends on the **threshold**.
- Explains why some runs look “worse” at 0.5 even with strong AUC.



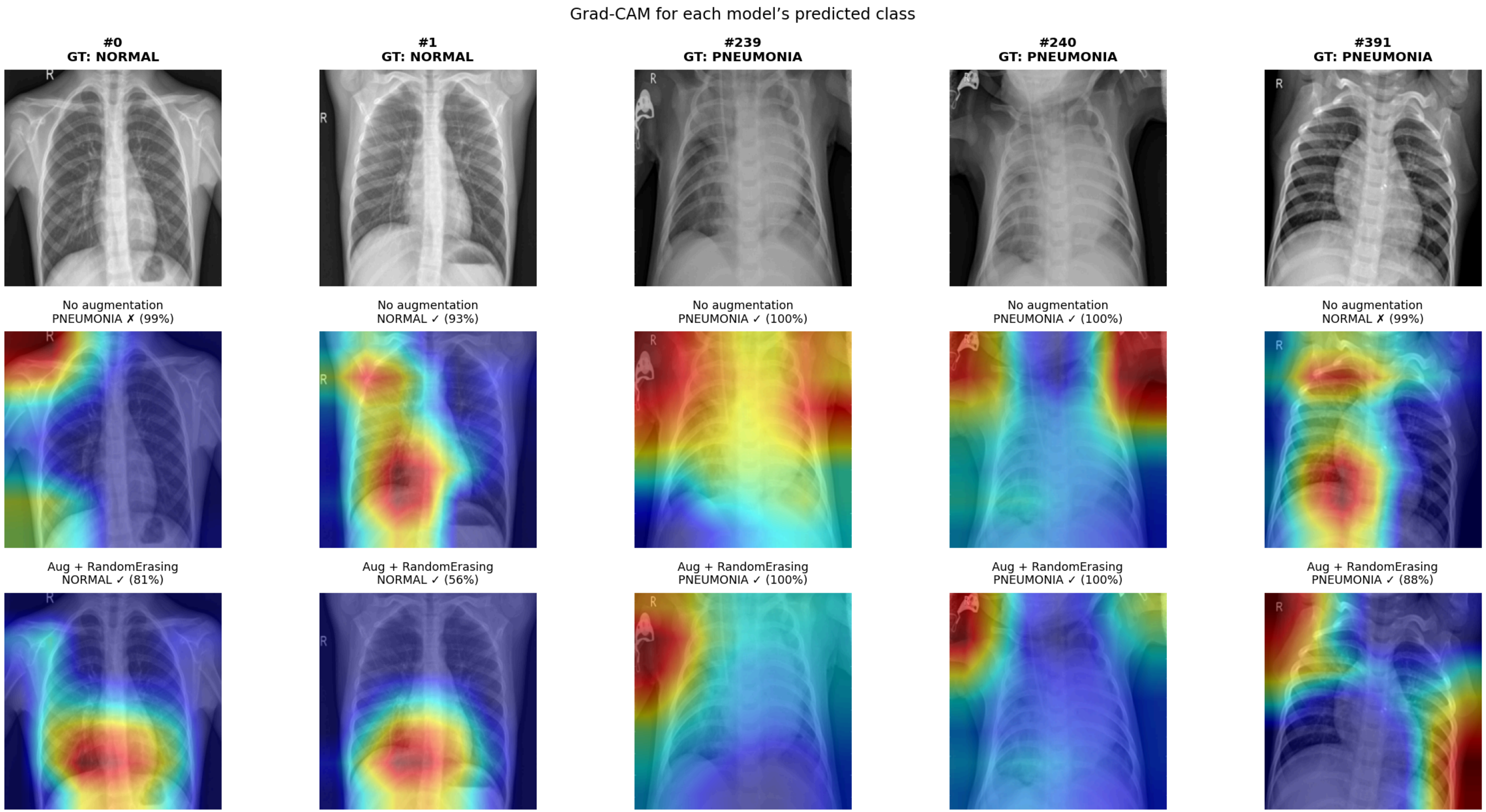
Qualitative analysis (Grad-CAM)

Goal: inspect *where* models look (explainability), not prove augmentation “fixes” attention.

Protocol:

- Fixed test indices → `configs/gradcam_manifest.json` (TP / TN / FP / FN examples)
- Grad-CAM targets **each model’s predicted class**
- Compare **aug off vs aug on** (same images): `results/gradcam/aug_off_vs_on.png`

Qualitative — Grad-CAM (aug off vs on)



Qualitative — what we observed

- Maps often highlight **lung fields and borders** — not always a focal consolidation
- **Aug off vs aug on:** on several cases, CAM looks **more central / anatomical** (less “random” edge focus)
- **Quantitative** changes (AUC, FP rate at 0.5) were **larger than visual** differences
- **Honest conclusion:** saliency is **illustrative**; needs external validation + better methods for clinical trust

Limitations & ethics

- Single public dataset · risk of **shortcuts** (view, edges, equipment)
- **No threshold tuning** on validation for clinical operating points
- **No external site** validation · high AUC ≠ safe deployment
- **Automation bias** if users over-trust scores
- **False negatives** remain clinically critical — report FN counts, not only AUC

Reproducibility

```
bash scripts/setup.sh && source .venv/bin/activate
bash scripts/reproduce.sh all
SEEDS="0 1 2" bash scripts/reproduce.sh robustness
```

Artifacts: `results/test_summary.csv`, per-run plots under `results/<stage>/seed_*/`, checkpoints in `checkpoints/`

Conclusion

1. Task is **feasible**; frozen ImageNet features are already strong (**AUC ≈ 0.95**).
2. Fine-tuning + aug gives **incremental** ranking gains; **not** a large leap.
3. **Focal loss** did not clearly beat CE in our runs.
4. **Qualitative Grad-CAM** shows **mixed** behavior — several cases look **more central/anatomical** with augmentation, but this is not a replacement for quantitative evaluation.
5. **Future work:** threshold tuning, external validation, richer pathology labels (e.g. multi-label CXR).

Thank you

Questions?

Repo: `MEDIMG-FS26-Pneumonia-Detection`